

You have 40 minutes to complete 10 problems. This is an individual round.

There is no penalty for guessing. Answer submissions are only valid if they are written on the answer sheet below.

First and last name			
p	Answer	p	Answer
1		6	
2		7	
3		8	
4		9	
5		10	

Problem 1. Let $M = 2$, $N = 3$, $Y = 5$, and $O = 7$. Compute $M \cdot N \cdot Y \cdot M \cdot O$.

Problem 2. Anna flips a fair coin 4 times. What is the probability it comes up heads every time? Express your answer as a common fraction in simplest form.

Problem 3. The side lengths of right triangle ABC are all positive integers less than 16. What is the sum of all possible lengths of the hypotenuse of ABC ?

Problem 4. Let square $ABCD$ be inscribed in circle ω_1 , and let circle ω_2 be inscribed in $ABCD$. Given that $AB = 2$, what is the difference between the areas of ω_1 and ω_2 ?

Problem 5. $10!$ can be written as $2^a 3^b 5^c 7^d$. What is $a + b + c + d$?

Problem 6. Let p and q be the roots of the quadratic $2x^2 - 7x + 3$. What is $p^2 + q^2$? Express your answer as a common fraction in simplest form.

Problem 7. Earl and Frank are playing a game. They each roll a fair 6-sided die. Earl wins if the product of their rolls has an odd number of divisors; Frank wins otherwise. If they play 36 independent games, what is the expected number of games Earl will win?

Problem 8. Angie, Alice, and Sneha each pick 3 distinct positive integers less than 12. Sneha knows Angie's number, but nobody else knows the other numbers that are chosen. Sneha claims that her number to the power of Angie's number ends in 3. Upon hearing this, Alice claims that her number to the power of Angie's number ends in 7. What is the sum of all possible products of the 3 numbers?

Problem 9. Cyclic quadrilateral $ABCD$ is inscribed in $\odot O$ with AC as a diameter. $\overline{BC}$ and $\overline{AD}$ are extended past B and A , respectively, intersecting outside the circle at E . Given that $AB = AD = 21$ and $BE = 72$, find the length of BD . Express your answer as a common fraction in simplest form.

Problem 10. A group of eight distinct friends bought a cheese pizza and pepperoni pizza, each with eight slices. The slices are divvied up among them so that each friend gets two slices. Assuming slices with the same topping are indistinguishable, in how many ways can this happen?